律的档位迁移与结构再生：SMPDL 方法论体系及结构生成盲重建首轮试验

Gear Migration and Structural Regeneration: The SMPDL Methodology and the First-Round Structure-Generation Blind-Rebuild Experiment

体系：SMPDL（Simulated Mathematical Practice with Dual-Lock）· G 系走廊 · 预注册工单 v1.3（J2 冻结）
数据基座：H1–H43 律伴随组库 + G1 试运行 + G 系正式轮三靶 12 候选 · Zenodo v3.6 密封（DOI 10.5281/zenodo.23599041）
执行窗口：2026-10-23 21:28 → 2026-10-24 · 设计：咖啡 · 协调：史努比（资格分离，禁判禁生成）· 版式：体系 F 纯白 v2

TL;DR — 本文合并三件为一：（一）SMPDL 方法论总纲——律学三维分类、生命周期、移除-涌现理论与 24 槽成分体系，及其五条历史实验线的实证基础（39 组 114 份案例库 + 回验十轮（前向 6/6）+ B4 三轮 LLM 盲判 + G-B1 62 题三层审查 + S8a 协议收敛）；（二）数学结构理论详细解释——三靶 24 槽解剖与结果的结构学解读；（三）G 系正式轮完整试验——预注册三区隔离下 12 候选盲重建的判定、对照与回验。核心发现一句话：**年代锁下的盲重建稳定抵达历史锚前一站（② 命中 83%）而锚核心零抵达；停点与错误形态双重贴合历史时序；人类出题 / 机器解题的分工画像获首轮实证。**

第一部分 · 中文完整版（一轮 v1.1 全文 + 第二轮增补章）

摘要

SMPDL（双锁模拟数学实践）把「数学结构的发现过程」变成可测量对象。本文首先给出其方法论体系：每条数学律携带三维身份证（强度分级 × 认识论档位 × 功能类型），律的档位迁移螺旋（E_0 默认 → I/D 质疑 → E′ 再定理化）构成读数学史的主叙事；移除-涌现理论（α 补偿 / β 解禁 / γ 回收三型 + γ 五形态 + 两阶段定律）刻画结构再生的机制；24 槽成分词汇表与 A1 载体 / A2 算子二分把「分解一个数学结构」变成标准查表作业。在此体系上，我们执行了 G 系正式轮：在显式知识年代锁（T-lock）与免联网条件下，让 6 个商用 LLM 以「同代数学家」身份对三个被拆除关键结构的领域（G3' 撤 Desargues 命题锁 1899 / G4 撤 T_2 分离公理锁 1914 / G5 撤变量绑定锁 1924 前）完成盲重建，经预注册的四判据时代忠实门（PFE）、四维机械评分卡、联网后见者四格判定（每靶 4 判定者）、一次性盲态微假说回验与五组对照臂检验。结果：12 候选中 10 个（83%）盲命中 T_L 锚后 T_S 锚前的真实历史结构（Hall 三元环骨架 / Kolmogorov 商与 US 层 / Tarski 柱状代数算子族），而三个锚核心全部零抵达（S/K 双构件基、Moulton 显式模型、Hochster 谱表征各 0/16）——「抵达锚前一站、不抵达锚核心」三靶一致；盲重建连错误形态都贴合历史时序；预测族 P1–P8 终局 6 过 2 半过，其中 P4（AI 取靶签名：具名 100% / 拆律 0%）与 P7（AI 补完动作与人类历史动作同格，四轮独立复现）联合成立，解锁「人类出题 / 机器解题」分工画像；退化基线 B1 实测暴露审查门对「换名请回」的正交失明，由评分卡非平凡性维度兜底。全部结论限定为「显式知识年代锁下的盲会话重建」首轮方向性记录。

§1 SMPDL 方法论体系

1.1 框架底座：三层两维

一切分析在 S/M/P × D/L 网格中进行：**S 语义层**（概念含义与命名——同名是否异义）、**M 数学层**（定理群与证明骨架——承重墙在哪）、**P 编程层**（形式化与计算可及性——库/算法是否拦截）；两维为**依存维 D**（前驱是否在场）与**逻辑结构维 L**（推理链是否合法）。三层联合断言：跨层结论只有同时通过 S/M/P 三层检查才成立——单层「看起来没问题」是全库反复出现的假象来源。

1.2 律学三维：律的身份证

第一维 · 强度分级：甲级（公理级，库内 11 条：交换/结合/分配/LEM/AC/DC/二阶归纳/平行公设×2/Archimedes/幂等）｜乙级（表征级，12 条：律 ⇔ 结构表征定理，拔它则伴随结构同生共死——「结构指纹」）｜丙级（规则级，2 条：intro/elim 规则对、收缩/弱化）。

第二维 · 认识论档位：E 档（定理律——数学内证明可裁，审计风险低）｜D 档（定义律——概念选择裁定）｜I 档（独立律——独立性证明裁定，**错用静默**，审计风险最高）。

第三维 · 功能类型：T1 定义律（对象构成——「这个对象是什么」）｜T2 保证律（行为构成——「过程如何运行」）｜T3 禁闭律（不变量——「什么保持不变」）。

档位迁移螺旋（读数学史的主叙事）：降档 E_0 默认 → I/D 质疑（平行公设 1868、交换律 1843、LEM 1930s、AC 1938/1963）；升档 I → E′ 再定理化（Gauss–Bonnet 回收平行公设后果、Zorn 相对化回报、bang 收缩律相对化回收（旧称「定价」，随语言修订注记对齐））。审计风险随 E<D<I 递增；同律两侧可异档；档位迁移是高产地平线（启发式表述，已从「定律」降格）。

1.3 律的生命周期

七阶段：L1 潜默 → L2 命名默认 → L3 边界摩擦 → L4 显性质疑 → L5 准独立 → L6 档位裁决 → L7 后续演化。历史结构：四大质疑集群（1843 代数解放 / 1899–1908 几何-集合 / 1931–38 元数学 / 1963–67 forcing）间隔约 30 年且工具先于质疑；裁决工具六代（几何模型 → 独立性方法 → 算术化 → forcing → 语义发明 → 构造性类型论+证明助手，进行中）；L3 长停留是常态——「意识到需要独立性证明比证明它更难」。

1.4 移除-涌现理论：结构再生的机制

三型涌现：α 补偿型（「谁顶班」——新结构接替旧律职责，Hahn–Banach/bang/正交模律）｜β 例外型（「被解禁」——被遮蔽对象族出现，四元数/例外李代数/平面 zoo）｜γ 统一回收型（「被回收」——旧律降格为新母结构特例，Gauss–Bonnet/概形）。

两阶段定律（前向 6/6 验证）：阶段一（移除时刻）涌现类型 ← 功能类型：T1→β、T2→α+元层、T3→β-现象；阶段二（回收时刻）γ ← 领域成熟度（与功能类型正交）。

γ 五形态：表征型（即时，乙级律天然自带）｜因果型（为修复而发明）｜渐近型（极限影子）｜成熟型（领域成熟后统一）｜翻译型（**失效模式本身对象化**——关键对/特殊化序，「病即信息」）。

方法论金句：伤口大小决定新生大小（乙级律移除生成学科 ×3）｜定价优于禁止＝相对化优于禁止（bang/Tonelli/AC 三连：被删保证仅在显式结构 D 内恢复有效）｜当对象背叛你，发明更大的结构（Dedekind/Lebesgue/Sobolev）｜减法倾向生结构、加法倾向生定理（审计降格为组织性观察）。

1.5 24 槽成分体系与 A1/A2 二分

最简结构判定：载体+算子+律束各一即成（magma 3 槽）；此后每加一个成分 = 一次升档。24 槽分五大类：A 骨架 7（载体/算子/律束/残存律/免费区/降档阶梯/镜像字典）｜B 定理 4（定理群/表征定理/补偿判据族/回收定理）｜C 对象 2（被禁族/病理族）｜D 过程 4（生命周期/档位迁移/锚点/迁移类型）｜E 涌现 5（对象/补偿结构/元层算法/跨组规律/构造性成本）。

A1 载体大类（场侧） vs **A2 算子大类（机器侧）**：「这个世界由什么组成」vs「在场上能做什么」；桥接类（定理层）把两侧钉在一起。**拆法谱系**：载体拆（换/弱化/升格载体）与算子拆（拆契约=律靶、拆机器=算子靶）。同构回声：泛代数「代数=(A;F)」、Bourbaki 母结构、范畴论对象/态射皆为此二分的变体。

§ 1.6 体系实证基础：SMPDL 历史轮次已跑数据（G 系之前）

SMPDL 方法论并非为 G 系临时搭建——其全部构件均由 2026-10 月内五条实验线的历史数据逐一锤炼而成。以下为各线实测记录（原始归档：`SMPDL_GitHub_Pages/reports/` 及父目录）。

（一）案例库与回验十轮（方法论的预测有效力）

资产	规模与实测
案例库	H1–H39 · 39 组 114 份报告（单体 158 + 对比 46 + 元文档 10）；律伴随组 25 组（64%）：甲级 11 / 乙级 12 / 丙级 2
回验轮次	**追溯 ×3**（H27/H28/H29 框架外对照：α+β 双主 / β 主+α 辅 / 零涌现——加法对照组零涌现 ✓）＋ **后验 ×4**（H30–H33：✅/❌→修订案 B/✅/✅——唯一偏差产出两个新概念并触发判据重构，「错误产生信息」）＋ **前向 ×6**（H34–H39：**6/6 全中**，含 bang 教科书命中、即时 γ、因果 γ、渐近 γ）
跨域锚	6 个：$\mathbb{Q}$-$\mathbb{R}$-*$\mathbb{R}$ 三角 / 动力三律轴 / no-cloning（H27×H36）/ 几何-代数镜像 / 交换代数三律轴 / 半环双律轴
学科级涌现	3 例（凸分析 / Sobolev-分布 / 均衡理论——全部为双面律，支持「学科级涌现 ⇔ 双面律」假说）
机器验证	Cabello KS 穷举 + 奇偶双确认；Mathlib 档位证据表（档位-形式化一致性 12/13）；Zenodo v2.7 在线（DOI 10.5281/zenodo.23355872）

（二）B4 三轮 LLM 档位盲判（判据可迁移性的精确划界）

让 LLM 对 H34–H43 前向 10 组盲判三问（Q1 功能类型 / Q2 涌现主型 / Q3 γ 回收方式），对照人工基准：

指标	R1 试点（半盲）	R2 正式（裸判）	R3 判据注入	裁决
Q1 功能类型	30%	70%	**90%**	判据文档化后过 80% 线 → **允许上机**（判据随脚本分发）
Q2 涌现主型	70%	**90%**	70%	裸判即高位 → **机器预填 + 人工复核**（R3 回落 = 过规则化副作用）
Q3 γ 细分	60%	30%	30%	**注入定义亦不可救** → 人工必填（区分依赖案例库 know-how，非 know-that）

三轮一致率均值 ≈53% → ≈63.3% → ≈63.3%：LLM 判据迁移落在 few-shot 带，但**分问分层后机器化方案清晰**（Q1 文档化上机 + Q2 裸机预填 + Q3 人工）。失败模式解剖两条：Q2 回落 = 判据注入须同时注入向量制（主型+辅型），否则规则刚性化反噬 20 点；Q3 顽固 30% = γ 字段应从案例库 RAG 检索而非定义生成——为 G 系「双审 + 判定者联网核验」的设计提供了直接前驱证据。

（三）G-B1 三层证明审查基准（S/M/P 联合审计的实测样本）

62 道数学证明（M001–M062）三层独立审查，层间不共享结论；P 层全部断言由 `verify_bench.py` 实际运行验证：

层	flag	pass	典型失败类
S 语言层	26	36	隐喻字面化（M001「结构上同构」）/ 同名异义（M002 Grandi 级数「和」）/ 图形依赖（M012「由图可见」）/ 前提静默沿用（M014）/ 循环论证（M017）/ 量词违规（M036 可数验证→全称断言）
P 编程层	25（含数值反例命中）	39（含 2 条 uncomputable 如实不猜）	反例直接命中：M004 缺素数前提（262 组反例）/ M009 半群无逆元 / M013 $(A+B)^2 \neq A^2+2AB+B^2$ / M017 Carmichael 数 561 / M025 递减列方向滑移

方法论意义：M 层「看起来没问题」的证明在 S/P 两层各被击落约四成——三层联合断言（§ 1.1）不是口号而是 62 题实测；「启发式泄漏」家族（数值外推/类比平移/显然化）是 LLM 生成证明的最高频病灶，直接催生 G 系 PFE 门的「概要级闭合口径 + 诚实降级条款」设计。

（四）S8a 协议收敛验证 + S8b 新律前向（判定协议的压力测试）

S8a（H34 Archimedes 律，协议 v1→v2）：v1 三系三档分裂（GLM ✅ / Kimi ✅ / DeepSeek 🟠）→ v2 双臂（Kimi 桌面端 J-D ✅ / DeepSeek J-E ✅）**全员同档收敛**——分裂臂在协议 v2 下翻转收敛，归因机械定位：D2 锚型声明制把「击破锚自由读法」（约 1900 前后 → 显影间隔 ≈60 年 ❌）钉死在 1891 Veronese（间隔坍缩 0–2 年）。残余自由度如实登记（显影年取 Hensel 1897/Hahn 1907 则 D2 翻 ❌）。

S8b（新律前向双轮，预测冻结后判定在审稿期）：R1 叠次交换律族（Moore–Osgood 家族，击破锚 1897 Pringsheim，T2 → 预测 α+元层）；R2 正则性分层律（椭圆正则性范式，击破锚 Spencer 1950s/显影 1963 Kohn，T2 → 预测 α+元层 + 预注册衍生律命名「次椭圆性」）。两轮预测均由 T→主型映射**机械查表产出**（零叙事自由度），序列墙隔离（永不并入 B4/前向池）。

与 G 系的衔接：S8 系证明「判定协议的可收敛性」与「查表预测的机械性」，G 系正式轮的判定提示词（锚型声明制/三维对照/四格分类）正是其直系后代；G-B1 的失败模式家族直接预演了 G 系 PFE 四判据的分工（词汇/动机/机制/可证性）。

§ 2 三靶结构解剖（数学结构理论详细解释）

2.1 三靶律身份证与结构画像

靶	拆除物	律身份证（三维）	历史涌现主型	关键结构锚
G3' H37	Desargues 命题 D	乙级表征 × $E_0 \to I \to E'$ 完整螺旋 × T1	β（非 D 平面解禁）+ γ-表征	Hilbert 1899 → Hall 1943（PTR + 归约定理）
G4 H41	T_2 分离公理	乙级表征 × $E_0 \to I \to E'$ 进行中 × T2	α+元层（US/商/网滤）+ γ-翻译	Hausdorff 1914 → Hochster 1969（谱表征）
G5	变量绑定机制	丙级规则 × $E_0 \to I \to E'$ 完整螺旋 × T2	α（S/K）+ β（组合子）+ γ-表征	Schönfinkel 1924（U/V=S/K + 组合完备性）

2.2 G3' 二十四槽要点（射影平面 · 三元环）

残存律：关联公理全体、局部调和构造。**免费区**：纯关联定理、对偶原理。**镜像字典**：D⇔结合律线性化；四点形对边点共线⇔1⊕1=0（Fano/特征 2）。**表征定理**：D ⟺ PTR 线性化为除环（乘法可不交换）。**补偿判据族**：Hall 五条三元环公理——以三元合成 T(m,p,q) 顶班二元乘加预设（α 型历史标本）。**回收**：γ-表征型即时（D 从公理变定理）。**被禁族**：非 Desargues 平面 zoo（Moulton）。**降档阶梯**：PTR → 线性 PTR（D 恢复）→ 除环 → 域（Pappus）。**档位螺旋**：D 走完 $E_0 \to I$（1899–1902）$\to E'$（VW 1907/Hall 1943）全程。**构造性成本**：非 D 模型显式构造成本——盲候选 12/12 诚实留白处。

2.3 G4 二十四槽要点（一般拓扑 · 谱空间）

免费区：连续像紧致、紧空间闭子集紧、连通性——不需分离的全免费带（候选 C4 的存活清单正是此槽盲重填）。**降档阶梯**：$T_2 \to$ US（序列唯一）$\to T_1 \to T_0 \to$ 无分离；Hochster 反向：谱空间 = 紧 T_0 + 特殊化序 + sober。**镜像字典**：分离公理 ⇔ 收敛行为（第一可数下 $T_2 \Leftrightarrow$US；$T_1 \Leftrightarrow$单点闭；$T_0 \Leftrightarrow$可辨）。**补偿判据族**：US/KC 中间层 + Kolmogorov 商——候选盲重建主战场（C2/C3 独立造出 US 层）。**回收**：γ-翻译型（多极限对象化为极限集/归宿集 L(s)——「病即信息」）+ γ-渐近型（特殊化序把坏拓扑回收为代数几何默认舞台）。**元层算法**：网/滤子（关于序列的元层判定工具）——候选诚实指认缺口处。跨组规律：「伤口大小决定新生大小」标本（乙级大承重缺口 → 生成整个非 Hausdorff 世界）。

2.4 G5 二十四槽要点（谓词逻辑 · 组合逻辑）

被拆机器：变量绑定（约束变元/辖域/代入/捕获避免）。**免费区**：单称命题区、有限合取析取区（候选 C3 核心观察）。**镜像字典**：量词⇔算子代数（$U_i/E_i \Leftrightarrow \forall/\exists$）；变元接线⇔参数置换/对角化（C/W ⇔ 换位/自复）。**补偿判据族**：位算子系统（量化代数化）与整体断言算子（量词原语化）两路。**回收**：γ-表征型（一阶逻辑 = 柱状代数方程理论）+ γ-翻译型（λ-项⇔组合子项）。**降档阶梯**：全量词语言 → 一元量词语言 → 无变元语言；Schönfinkel 反向：多元 → 一元连续作用（currying）——降档与升格同一枚硬币。**病理族**：复现模式无通用有限基（W/T_f 特例覆盖）；二阶墙（概念遍历不可表达——自建语言边界，历史补漏=加类型层）。

2.5 结果的结构学解读（理论 × 试验对读）

1. **锚前一站收敛律 = A6 降档阶梯的单步移动**：三靶盲候选抵达的都是阶梯上离拆除物最近的下一档（三元合成 / US 层 / 逐坐标算子）——盲会话能做档位内移动（保守范式延伸的极限 = 保留旧范式最多前提下的最小修改），不能做 D4 框架迁移（范式转换）。锚核心全部需要换桥接类填法，超出单步能力。
2. **错误保真 = 档位迁移的认知顺序**：候选「D⇔结合律」过强断言 = 1907 年 VW 分层前的 E_0 默认态错觉（律=结构未分层直觉）；候选沿生命周期 L4（感到伤口）分布、停在 L5 门前——复现的不是知识缺口而是认知顺序。
3. **H-A2 证伪 = 槽位就近回填原理**：拆算子后涌现「算子的对象化版本」（位算子系统）而非仅向判据层逃逸——涌现以被拆槽位为原料重建，锚侧与涌现物同型。反馈结构理论：A2/C1 边界（机器 vs 机器构成的对象族）比预想更可渗透（E1-h 修订）。
4. **P7 同格 = 共享 24 槽深层语法**：LLM 与人类历史共享同一张填充语法；分工界面切在 A3 律束槽的「选择动手术」（人类问题意识）与「执行手术」（机器执行语法）之间——P4 显示 AI 从不自发选择拆律（0%），P7 显示被置于拆律情境后动作与人类同格。
5. **B1 部分抓获 = 等价复现不是涌现**：换名请回填满修复槽但非平凡性为 0——真重建必须出现 α/β/γ 之一机制，等价复现在 24 槽表上不是涌现（无新槽位填法）。E1-j 条款本质是把涌现判据提升到门的明文。
6. **三靶涌现同型性**：历史涌现主型与盲候选涌现主型全部同型（G5 换到柱状代数平行线但仍是 α-代数化）——「阶段一律律」（T 定涌现主型）在盲条件下依然成立，本试验对其最强支持；G5 换线测出「同型内多实现路径」的谱系宽度。

§ 3 试验设计（G 系正式轮）

3.1 三区知识分区与预注册

三区制：A 区（现代知识合法：取靶/冻结件撰写）→ B 区（T-lock 无前视：生成/扫描/审查）→ C 区（轮后集中：判定/评分/回验），区间以 git commit 单向门隔离，B 区产物 C 区永不回改。**资格分离**：协调方禁判禁生成；生成/审查/判定各为独立会话；判定者盲于审查轮。**预注册**：全部判据、权重（修复度 40% / PFE 可证性 25% / 自洽性 20% / 非平凡性 15%）、预测族 P1–P8、微假说 H-A1/A2 随工单 v1.3 commit 冻结。

3.2 三靶与 B 区流程

靶	拆面	双锚（T_L / T_S）	锚距	候选
G3'	律（机器契约）	Hilbert 1899 / Hall 1943	44 年	4 厂牌（Gemini-Flash/DeepSeek/Kimi/ChatGPT-Flash）
G4	结构（场约束）	Hausdorff 1914 / Hochster 1969	55 年	4 厂牌（Gemini/DeepSeek/Kimi/ChatGPT）
G5	算子（机器本体）	1924 前默认 / Schönfinkel 1924	0 年	4 厂牌（Gemini-Flash/DeepSeek/Kimi/ChatGPT-免登）

B 区流程：生成（免联网+自报+禁读描述性文件）→ 年代机械扫描（🟡旁注/🔴定义核心分级）→ PFE 四判据门（词汇/动机/机制/可证性；诚实降级保护；只判不改建议；门分级不淘汰）。每靶双异厂审查者；C 区判定每靶 4 判定者联网核验锚论文，按四格分类（① T_L 前命中 / ② T_S 后命中 🔥 ⩾2 规则级同构点 / ③ 新颖 / ④ 违规）。

§4 结果

4.1 PFE 门与评分卡

12 候选 = **5 完整结构 + 6 方向性草案 + 1 双审分歧**（草案全保留进判定，G-L5）。待裁项闭合：滤子 🔴 / 结构组合子 🔴 / W・C 记号层 🔴（异议留档）。评分卡：G3' 胜者 C3 DeepSeek（1.00 唯一）；G4/G5 各三方并列 1.00（卡顶饱和发现，E1-c）。G1 回溯校准（D-18）：机械排序与预期一致 → **权重校准通过**。盲审纪律旁证：DeepSeek 判自厂候选 FAIL、Kimi 判自厂候选 FAIL——无自审豁免。

4.2 判定总表与三靶不变量

靶	①	② 🔥	③	④	最强命中
G3'	—	C1/C3/C4 四方一致 + C2 分裂②×2	C2 ③×2	—	C3 规则级（≈Hall PTR 逐条同构）；C4 归约级（VW 1907 核心条款双盲命中）
G4	1 票自矛盾	C1–C4 主判	C4 1 票异议	—	C2（Kolmogorov 商 + 第一可数 $T_2 \Leftrightarrow$US 盲证 + 余可数反例；三家独立排第一）
G5	C3 主判 2/4	C2/C4 四方一致	C1 分裂 2 票	—	C2/C4（柱状代数平行线：U_i↔柱化 c_i 精确对应）

> **三靶不变量**：② 主判 10/12（83%）；三个锚核心全部零盲抵达（S/K 0/16、Moulton 0/16、Hochster 0/16）而锚前一站全部多人抵达——「**抵达锚前一站、不抵达锚核心**」。双构件旗标 0/16：G5 四候选全部流入量化代数化线、零流入组合子线。

4.3 微假说回验（一次性盲态）与对照臂

H-A1（T 定涌现主型）：✅ **支持**——T1 靶涌现对象族 / T2 靶涌现判据元层，三靶一致。**H-A2**（锚侧定落位）：⚠️ **载体侧全中；算子侧严格版证伪**——G5 双落位（算子对象化+元层），涌现物与锚同型 → E1-h 细化「锚侧→涌现物模态」。

#	预测	结果	裁定
P1	判据区分力 H≥80% vs R≤40%	88% vs 10%	✅
P2	R 失败集中于平凡+致命	平凡 13/13 单侧（抽样框偏差）	⚠️
P3	H 聚带宽 vs R 双尾	成立	✅
P4	AI 签名：具名≥90% 且拆律≈0	100% / 0%（46 族）	✅
P5	认可↔验证器正相关	86% vs 33%（n=31 降档）	✅ **方向性**
P6	六型目录完备 25/25	型外 0	✅
P7	AI 动作与人类同格	四轮独立复现	✅
P8	退化基线被拦截	B0 ✅ / B1 部分抓获	⚠️

> **P4 ∧ P7 联合成立 → 「人机分工画像：人类出题 / 机器解题」正式解锁**（D-21 预注册条件满足）。

4.4 B1 退化基线实测

夹具 = T_2 换名「双邻域相容条件」逐字请回（词汇层无懈可击）。三盲审者 PFE-1/2/3 全 PASS、**零等价复现指认**（ChatGPT 独家 PFE-4 FAIL 于断言过强）；机械评分卡：非平凡性 **0.00**（唯一抓 B1 的维度）、修复度 1.00 陷阱不采信、加权 0.18 < 真候选地板 0.25。**结构性发现：PFE 门的判据轴（时代忠实）与作弊轴（等价复现）正交**——复现原条件恰是最 era-legal 的内容；兜底 = 非平凡性维度（15% 权重的意外英雄）→ E1-j 高优先。

§5 分析：三大发现

1. **保守范式延伸吸引子**（三靶交叉验证）：盲修复直觉从现有范式的可运算化延伸出发——撤 D → 坐标运算整体化（保留「坐标化」范式）；撤 T_2 → 收敛行为分层（保留「邻域-序列」范式）；撤变元 → 量词代数化（保留「关系代数」范式）。三个历史锚核心全部需要范式转换，无一被盲触发。「抵达深度跟任务内在难度走、不跟锚距走」——G5 锚距 0 年反而未及 S/K。
2. **停点 + 错误双重历史保真**：候选非 D 模型零构造全诚实留白 = 1899 真实同代人状态（Hilbert 存在性论证 → Moulton 1902 显式模型）；候选错误 = 1907 前民间状态 / 前规范期术语混乱 / 1920s 谱系同款蕴含混乱。错误类型按时序分布——盲重建「走在历史的路上」。历史补漏比照：候选诚实指认的边界恰是历史实际越过的边界，且越过方式与 MR 建议一致。
3. **双轴解耦与分工画像**：忠实（PFE）× 质量（MR）三靶三模式（同冠/解耦/偏移）——「忠实于时代」与「逼近历史答案」可分离；P4∧P7 给出分工画像：AI 100% 具名取靶、0% 拆律，但被置于拆律情境后动作与人类四轮同格——人类出题 / 机器解题。F7（诚实未决申报，旧称「诚实悬置」）为 AI 独有行为候选（人类史未见以未决申报为发表动作者，本试验 16/16 候选共现诚实降级层）。

§6 诚实披露与局限（八条）

1. **参数污染不可排除**：全部结论限定表述为「显式知识年代锁下的盲会话重建」，不声称「无知条件下的独立发现」；三靶不变量（锚核心零抵达）为该表述提供内部证据——若前视不受抑制，锚核心应如锚前一站一样被「回忆」出来。
2. **锚年 = 史学判断**（异议附记制；判定中两处史实校正已入 E1-f：Moulton 限定、Fano 溯源、Schönfinkel U≠S）。
3. **操作化旗标**：评分卡维度→公式未微预注册（协调方代拟透明登记）；卡顶饱和；G1 校准证据层级降档；PFE-4 诚实降级缺口；单字母记号判据优先序；② 门槛概念级二分（三靶三复现 E1-g）。
4. **样本量**：靶级 n=3(+1)、候选级 n=12——首轮方向性记录，不外推；防挖掘三锁下不做事后子集分析。
5. **M8 验证器自警**：PFE+判定侧本身是筛选——不可形式化靶系统性缺席；B1 实测正交失明加深此条。

6. **判定者侧边界**：结构命中 ≠ 优先权；判定者与候选厂牌同源（实测无自审豁免，仍全量登记）；一次未实时联网（降档）；一次厂牌内自不一致（按正文主判并档）。
7. **流程偏差**：G4 双审厂牌替换（被墙补位，追认）；超额审查者分列；B1 超单审配置——E4 双面台账全登记（正向：多审纪律两次实证防线价值）。
8. **MR 扩展轮地位**：C 区扩展非预注册，独立分列永不进算分（双轴防火墙）；v1.4 起瘦身为纯质量轴（E1-d）。

§ 7 结论

1. **能力**：年代锁下 LLM 盲重建稳定抵达真实历史的锚前一站（83% ② 命中）；锚核心零抵达——保守范式延伸吸引子三靶验证。
2. **路径**：停点与错误形态双重贴合历史时序——盲重建生态效度的结构性旁证。
3. **分工**：人类出题 / 机器解题画像获首轮实证（P4∧P7）；F7 诚实未决申报为 AI 独有行为候选。
4. **方法论**：三区隔离 + 预注册 + 分歧即数据的「受控历史重演」范式可复用；B1 给出验证体系纵深刻画；E1 登记十项（a–j）构成 v1.4 修订底稿。SMPDL 结构体系同时充当实验设计语言、结果解释语言与验证体系设计依据——一套词汇表贯穿三层用途，方法论自洽性首轮实证。

第二轮增补 · 2026-10-24（v2.0 并入）

§ R2-1 第二轮设计与执行

工单 **v1.4**（E1 十项 + G-L11/12 + P9 迭代臂预注册，commit `150ab55` 冻结）首轮全条款执行。三靶：**G6 交换律拆除**（Hamilton 1843 / Frobenius 1877 双锚）、**G7 相体积守恒拆除**（Poincaré 1890/1899，载体侧 × T3 六格唯一空格）、**G8 唯一因子分解拆除**（Kummer 1847 / Dedekind 1871 双面律）。Zone-B 12 候选（Zenodo v3.8 密封 DOI 10.5281/zenodo.23615249）→ Zone-C 全链（PFE 双审 6 + MR 纯质量轴 6 + 五档评分卡 + 双判联网判定 6 + 史学对照 + H-A2′ 回验 + 对照臂 + **P9 三臂**）。外部会话 22 个；协调方零生成（G-L1）。

§ R2-2 判定结果：锚核心首次命中

TL;DR：G8-C4「因束」经双判定者一致裁定为 **1871 理想论核心命中**（七行规则级对应表全同构：集化定义/运算/反演/素=极大/存在唯一/显式分解 (2)=P^2(3)=Q・Q′/非主性）——**G 系历史首次锚核心命中**。第一轮不变量「抵达锚前一站、不抵锚核心」在第二轮被打破，修正为**「骨架可抵、地基留缺」**：格点规范形引理（唯一未证支点）经裁定为 (a) 类——1847 工具箱原则上可补；ChatGPT 判定者增量发现所述条件 a ∣ b^2+5 不充分（ω 封闭迫使 a ∣ 2b）。

靶	结构命中（双判收敛层）	关键数字
G6 交换律	Hamilton 1843 锚答案 **4/4**；Frobenius 片段 1/4（C4）	0 年锚距可组合答案全抵（vs 一轮 G5 S/K 0/16）
G7 相体积守恒	C4 等距回复唯一收敛 ②（拓扑动力学平行线）	1899 几何层 0/4（预注册零抵达 ✅）；测度论方法革命不可通达
G8 UFD	**C4 因束 = 1871 核心命中**	与 Dedekind 1877 示范例逐项同物；γ 锚后站（任意 D/类群）零抵达

§ R2-3 三大理论更新

1. **不变量修正**：一轮「锚前一站不抵核心」→ 二轮「骨架可抵、地基留缺」——对象/定理结构层可盲复制（评分卡→MR→判定三层合流：锚核心级交付与证明缺口共变）。
2. **pivot 三档**（0 年锚距梯度细化）：零 pivot（四元数 = 任务已 prescribe）4/4 ＞ 对象革命（理想 = 任务半暗示）1/4 ＞ 方法革命（测度化 = 不可提示）0/4 直抵 + 1/4 绕道。答案所需概念革命的类型，而非年代距离，决定盲抵达率。
3. **H-A2′ 部分支持 + H-A2″ 候选**：算子侧 2/2 逐字兑现（拆算子契约 → 算子以对象形态重建 + 元层审计）；载体侧首个换线（G7 判据族涌现，对象型 0/4 双判一致）。根源分析：「拆构成律 → 新对象补偿；拆保证 → 新判据补偿」——**补偿物镜像拆除物形态**（H-A2″，事后构造，防挖掘三锁下三轮预注册后方可检验）。

§ R2-4 P9 同代评审迭代臂（三臂收官）

挂载 G8-C4（判定层已裁唯一缺口 = 规范形引理）。十步 Δ 阶梯（判定轮同尺 + γ 锚后站红线）+ 四计数器（撤回/新增/**修补**/诚实声明）。

臂	输入	Δ	撤/新/修/诚	β2 反演律终态
主链（Kimi→DeepSeek→ChatGPT）	真评审 7 项	**1.0**	2/1/4/4→8	增强半步（普遍性未决申报；定理宣称撤回）
安慰臂（Kimi）	零数学意见	0	1/0/0/4→5	半步原样
自评臂（Kimi）	自查 5 项	**1.5**	0/0/5/4→4	**全步（引理会话内完整证明）**

三发现：① **Cauchy 角命中**——评审 DeepSeek 在零判定层信息下自见 b1 引理缺口（Liouville 1847-03-01 当场质疑角色复现；且增量给出 r1 § 四.2 的正确反例 2+ω/2−ω）；② **修补第三模态确立**——试点与安慰臂皆撤回主导，真评审输入下主链转修补主导、自评臂全修补零撤回：「清创 vs 生长」二分不足，H-B 按修订者-构造关系分层（E1-n）；③ **自评 1.5 > 主链 1.0**（P9-2 对照臂预测部分证伪）——评审具体度 × 修订者身份双变量交互：外人修订保守撤回、家人续修直达补全（同厂知识连续性；E1-m 补格）。**结构-诚实张力**：同一 r1，自评臂「宣称落地+引理全」vs 主链「撤回宣称+引理悬」——最深者非最诚实（与评分卡稳健-深远解耦同型）。

§ R2-5 双面律与学科级涌现（第 4 数据点）

> 「学科级涌现 ⇔ 双面律（T1+T3）」假说第 4 数据点支持：G8 双面 → 全轮唯一锚核心命中；理想论历史上生成整个代数数论学科（已知正例）；盲样本最深抵达恰落此靶。双面律第 1–3 数据点（单面靶）均无学科级涌现。

§ R2-6 史学勘误（E1-f 附记制）

判定者联网核验产出：S-1 Kummer 1844 载于 Breslau 纪念文集（非 Crelle 35）；S-2《天体力学新方法》卷一 = 1892（Acta-1890 论文锚不变）；S-3 四元数日期 10-16（四候选统一 10-15 夜框架 = 训练数据平滑痕迹）；S-4 Frobenius Crelle 84 双重 1877/1878 标注；S-5 Kummer 信双时间线并档待复核。

§ R2-7 第二轮结论

第二轮在 v1.4 全条款（预注册评分公式、② 门槛二分、等价复现指认（旧称「复原指认」，随 2026-10-26 语言修订注记对齐）、era-scan 闸、迭代分列）下完成：锚核心零抵达不变量被打破并修正为「骨架可抵、地基留缺」；抵达率由 pivot 类型而非年代距离决定；同代评审机制部分复现真实历史（Cauchy 角命中 + Lamé 六周循环同构）并揭示修补第三模态与修订者身份效应。E1 v1.5 池（E1-k/l/m/n + S 勘误 + H-A2″ 三锁）登记待三轮。Zenodo v3.9 发布；序列墙维持。

Part II · English Full Version

Abstract

SMPDL (Simulated Mathematical Practice with Dual-Lock) turns the discovery process of mathematical structures into a measurable object. This paper merges three components: (i) the SMPDL methodology — the three-dimensional identity card of every mathematical law (strength grade × epistemic gear × function type), the gear-migration spiral (E_0 default → I/D challenge → E′ re-theorematization) as the master narrative for reading the history of mathematics, the removal–emergence theory (α compensation / β liberation / γ recycling, five γ-forms, two-stage law), and the 24-slot component taxonomy with the A1 carrier / A2 operator dichotomy; (ii) a detailed structural dissection of the three experimental targets on this framework; (iii) the G-series formal round — a preregistered, three-zone-isolated blind-rebuild experiment in which six commercial LLMs, placed under explicit knowledge date-locks (T-lock) with no network access, rebuilt three fields with key structures dismantled (G3′: Desargues proposition removed, locked 1899; G4: T_2 separation axiom removed, locked 1914; G5: variable binding removed, locked pre-1924). Twelve candidates passed a preregistered four-criterion period-faithfulness gate (PFE), a four-dimension mechanical scorecard, networked hindsight four-bin classification (four judges per target), a one-shot blind microhypothesis verification, and five control arms. Results: 10/12 candidates (83%) blindly hit real historical structures lying between the T_L and T_S anchors (Hall ternary-ring skeletons / Kolmogorov quotient & US layer / Tarski cylindric-algebra operator family), while all three anchor cores saw zero blind arrivals (S/K two-primitive basis, Moulton explicit model, Hochster spectral representation: 0/16 each) — a tri-target invariant: **reach the station before the anchor, never the anchor core**. Blind rebuilds reproduced even the error morphology of history. Prediction family P1–P8 closed at six passes and two partials; P4 (AI targeting signature: 100% named-conjecture, 0% law-dismantling) and P7 (AI completion actions same-lattice as human history, four independent replications) jointly unlocked the “humans pose / machines solve” division-of-labor portrait. Degenerate baseline B1 exposed the review gate's structural blindness to renamed restoration i.e. equivalent recurrence (orthogonal axes), backstopped by the scorecard's non-triviality dimension. All conclusions are bounded as first-round directional records of “blind-session rebuilding under explicit knowledge date-locks.”

§ 1 The SMPDL Methodology

1.1 Foundation: three layers, two dimensions

All analysis proceeds on an S/M/P × D/L grid: **Semantic layer** (meaning and naming — same name, different sense?), **Mathematical layer** (theorem groups and proof skeletons — where are the load-bearing walls?), **Programming layer** (formalizability and computational reach — do libraries intercept?). Two dimensions: **Dependency D** (are prerequisites present?) and **Logical structure L** (are inference chains legal?). Joint assertion: cross-layer conclusions require passing all three layers simultaneously.

1.2 The three-dimensional identity card of a law

Strength grade: Grade-A (axiomatic; 11 in library: commutativity, associativity, distributivity, LEM, AC, DC, second-order induction, parallel postulate ×2, Archimedes, idempotence) | Grade-B (representational; 12: law ⇔ structure representation theorem — "structural fingerprints"; disturbing one kills the whole companion structure) | Grade-C (rule-level; 2: intro/elim pairs, contraction/weakening).

Epistemic gear: E (theorem-law, adjudicated by proof; low audit risk) | D (definition-law, adjudicated by concept choice) | I (independent-law, adjudicated by independence proofs — **misuse is silent**, highest audit risk).

Function type: T1 definitional (what the object *is*) | T2 guarantee (how processes *run*) | T3 confinement (what stays *invariant*).

Gear-migration spiral: demotion E_0 default → I/D challenge (parallel postulate 1868, commutativity 1843, LEM 1930s, AC 1938/1963); promotion I → E′ re-theorematization (Gauss–Bonnet recovers parallel-postulate consequences; Zorn prices AC's return; bang prices contraction's return). Audit risk increases E<D<I; the same law can sit at different gears on two sides; gear migration is the productive horizon for reading mathematical history.

1.3 Life cycle of a law

Seven stages: L1 latent → L2 named-default → L3 boundary friction → L4 explicit challenge → L5 quasi-independence → L6 gear adjudication → L7 subsequent evolution. Historical texture: four challenge clusters (1843 algebraic liberation / 1899–1908 geometry–set theory / 1931–38 metamathematics / 1963–67 forcing) spaced ~30 years, tools preceding challenges; six generations of adjudication tools (geometric models → independence methods → arithmetization → forcing → semantic invention → constructive type theory + proof assistants, ongoing). Long L3 residence is the norm — realizing that an independence proof is needed is harder than proving it.

1.4 Removal–emergence theory

Three emergence types: α compensation ("who takes the shift" — Hahn–Banach, bang, orthomodular law) | β exception ("liberated" — quaternions, exceptional Lie algebras, plane zoo) | γ unification-recycling ("recycled" — Gauss–Bonnet, schemes).

Two-stage law (forward 6/6): stage one (at removal) emergence type ← function type (T1→β, T2→α+metalayer, T3→β-phenomena); stage two (at recycling) γ ← field maturity (orthogonal).

Five γ-forms: representational (immediate; Grade-B laws carry them natively) | causal (invented for repair) | asymptotic (limit shadows) | maturity (late unification) | translational (**the failure mode itself is objectified** — critical pairs, specialization order; "pathology is information").

Methodological maxims: wound size determines newborn size | pricing i.e. relativization beats prohibition (bang/Tonelli/AC) | when objects betray you, invent larger structures (Dedekind/Lebesgue/Sobolev) | subtraction breeds structure, addition breeds theorems (demoted to organizational observation).

1.5 The 24-slot taxonomy and the A1/A2 dichotomy

Minimal structure: one carrier + one operator + one law-bundle (a magma, 3 slots); each added component is one gear-up. Five classes: A skeleton (7: carrier, operator, law-bundle, residual laws, free zone, demotion ladder, mirror dictionary) | B theorems (4: theorem group, representation theorem, compensation criteria, recycling theorem) | C objects (2: forbidden families, pathology families) | D process (4: life cycle, gear migration, anchors, transfer types) | E emergence (5: objects, compensation structures, metalayer algorithms, cross-group regularities, constructive cost).

A1 carrier side (“what the world is made of”) vs **A2 operator side** (“what can be done on the field”), bridged by the theorem layer. Dismantling genealogy: carrier-side removals (swap/weaken/upgrade the field) vs operator-side removals (contract-targets vs machine-targets). Isomorphic echo: universal algebra's (A;F), Bourbaki's mother structures, category theory's object/morphism.

§ 1.6 Empirical Foundation: Previously Run SMPDL Data Lines (Pre-G-Series)

The SMPDL methodology was not assembled for the G series — every component was forged by five experimental lines during October 2026. Measured records (originals archived under `SMPDL_GitHub_Pages/reports/` and parent directory):

(1) Case library and ten verification rounds (predictive validity)

Asset	Scale & measured results
Case library	H1–H39: 39 groups, 114 reports (158 monolithic + 46 comparative + 10 meta); 25 law-companion groups (64%): Grade-A 11 / B 12 / C 2
Verification rounds	**Retrospective ×3** (framework-external controls: α+β dual-main / β-main+α-aux / zero emergence — addition control confirmed zero ✓) + **post-hoc ×4** (✅/❌→Revision-B/✅/✅ — the single miss produced two new concepts and triggered criterion reconstruction: “errors generate information”) + **forward ×6** (H34–H39: **6/6 all hits**, incl. bang textbook hit, immediate γ, causal γ, asymptotic γ)
Cross-domain anchors	6 (ℚ-ℝ-*ℝ triangle / dynamics triad / no-cloning / geometry-algebra mirror / commutative-algebra triad / semiring dual axis)
Discipline-level emergences	3 (convex analysis / Sobolev-distributions / equilibrium theory — all double-faced laws, supporting the “discipline-level emergence ⇔ double-faced law” hypothesis)
Machine verification	Cabello KS exhaustion + parity double-check; Mathlib gear-evidence table (gear-formalization consistency 12/13); Zenodo v2.7 online (DOI 10.5281/zenodo.23355872)

(2) B4 three-round LLM blind gear-judgment (precise boundary of criterion transferability)

LLMs blindly judged three questions (Q1 function type / Q2 emergence main-type / Q3 γ-recycling mode) for the H34–H43 forward ten groups, against frozen human baselines:

Metric	R1 pilot (semi-blind)	R2 official (bare)	R3 criteria-injected	Ruling
Q1 function type	30%	70%	**90%**	crosses the 80% line once criteria documented → **machine-admissible** (criteria ship with scripts)
Q2 emergence main-type	70%	**90%**	70%	high when bare → **machine pre-fill + human review** (R3 drop = over-regularization side-effect)
Q3 γ subdivision	60%	30%	30%	**injected definitions cannot save it** → mandatory human field (distinction requires case-library know-how, not know-that)

Three-round mean agreement ≈53% → ≈63.3% → ≈63.3%: LLM criterion transfer sits in the few-shot band, but **per-question stratification yields a clear mechanization plan** (Q1 documented + machine; Q2 bare machine pre-fill; Q3 human). Failure-mode dissection: the Q2 drop shows criteria injection must include the vector scheme (main+aux types) or rule-rigidity bites back 20 points; the stubborn Q3 30% shows γ fields should be retrieved from case libraries (RAG), not generated from definitions — direct precursor evidence for the G-series dual-review + networked-judge design.

(3) G-B1 three-layer proof-audit benchmark (S/M/P joint audit, measured)

62 mathematical proofs (M001–M062) under three independent audit layers with no cross-layer sharing; all P-layer assertions actually executed via `verify_bench.py`:

Layer	flag	pass	Representative failure classes
S semantic	26	36	metaphor literalization (M001 "structurally isomorphic") / same-name-different-sense (M002 Grandi series "sum") / diagram-dependence (M012 "as seen from the figure") / silent premise carryover (M014) / circular reasoning (M017) / quantifier violation (M036 countable-check→universal claim)
P programming	25 (numeric counterexamples hit)	39 (incl. 2 honestly-marked uncomputable)	counterexample hits: M004 missing primality premise (262 counterexamples) / M009 semigroup lacks inverses / M013 $(A+B)^2 \neq A^2+2AB+B^2$ / M017 Carmichael 561 / M025 monotonicity direction slip

Methodological meaning: proofs that "look fine" at the M layer fall at S/P layers at ~40% rates — the three-layer joint assertion (§ 1.1) is a 62-item measurement, not a slogan; the "heuristic leakage" family (numerical extrapolation / analogy transfer / obviousness-pleading) is the highest-frequency LLM-proof pathology, directly motivating the G-series PFE gate's sketch-level closure standard and honest-degradation clause.

(4) S8a protocol convergence + S8b new-law forward rounds (stress tests of the judgment protocol)

S8a (H34 Archimedes law, protocol v1→v2): v1 three-way three-tier split (GLM ✅ / Kimi ✅ / DeepSeek 🟡) → v2 two arms (Kimi-desktop J-D ✅ / DeepSeek J-E ✅) **full-tier convergence** — the split arm flipped under protocol v2; mechanical attribution: the D2 anchor-declaration regime pinned the free "breakthrough-anchor" reading (~1900 → ≈60-yr manifestation gap ❌) to 1891 Veronese (gap collapses to 0–2 yrs). Residual freedom honestly registered (manifestation year at Hensel 1897 / Hahn 1907 would flip D2 ❌).

S8b (new-law forward dual rounds, predictions frozen, judgments in review period): R1 iterated-interchange law family (Moore–Osgood family, breakthrough anchor Pringsheim 1897, T2 → predicted α+metalayer); R2 regularity-stratification law (elliptic regularity paradigm, breakthrough anchor Spencer 1950s / manifestation Kohn 1963, T2 → predicted α+metalayer + preregistered derived-law naming "subellipticity"). Both predictions were produced by **mechanical table lookup** (zero narrative freedom), sequence-walled from all prior tallies.

Bridge to the G series: S8 demonstrated the judgment protocol's convergence and the table-lookup predictions' mechanicality; the G-series judgment prompts (anchor-declaration regime / three-dimensional comparison / four-bin classification) are direct descendants; G-B1's failure-mode family directly rehearsed the PFE four-criterion division of labor (vocabulary / motivation / mechanism / provability).

§ 2 Structural Dissection of the Three Targets

2.1 Law identity cards

Target	Dismantled	Identity card	Historical emergence	Structural anchors
G3′	Desargues proposition D	Grade-B representational × full $E_0 \to I \to E'$ spiral × T1	β (non-D planes liberated) + γ-representational	Hilbert 1899 → Hall 1943 (PTR + reduction theorem)
G4	T_2 separation axiom	Grade-B × $E_0 \to I \to E'$ in progress × T2	α + metalayer (US/quotient/nets) + γ-translational	Hausdorff 1914 → Hochster 1969 (spectral duality)
G5	Variable binding	Grade-C rule × full spiral × T2	α (S/K) + β (combinators) + γ-representational	Schönfinkel 1924 (U/V=S/K + combinatory completeness)

2.2–2.4 Key slots per target (condensed)

G3′ — Free zone: pure incidence theorems, duality. Mirror dictionary: D ⇔ associativity-linearization; quadrangle diagonal collinearity ⇔ 1⊕1=0 (Fano/char-2). Compensation: Hall's five PTR axioms — the ternary composition T(m,p,q) replaces binary ±× presuppositions (the historical α specimen). Recycling: γ-representational (D becomes a theorem). Forbidden family: non-Desarguesian plane zoo. Demotion ladder: PTR → linear PTR → division ring → field. Constructive cost: explicit non-D models — where all 12 blind candidates honestly suspended.

G4 — Free zone: continuous images of compacta, closed subsets of compact spaces, connectedness (no separation needed — candidate C4's survival list blindly refilled this slot). Ladder: T_2 → US → T_1 → T_0; Hochster's reverse: spectral = compact T_0 + specialization order + sober. Mirror: separation ⇔ convergence behavior (first-countable T_2⇔US). Compensation: US/KC intermediate criteria + Kolmogorov quotient (the blind-rebuild battleground — C2/C3 independently reconstructed the US layer). Recycling: γ-translational (multi-limit objectified as limit sets L(s) — "pathology is information"). Metalayer: nets/filters — where candidates honestly identified the gap.

G5 — Dismantled machine: variable binding. Free zone: singular propositions, finite connectives. Mirror: quantifiers ⇔ operator algebra (U_i/E_i ⇔ ∀/∃); variable wiring ⇔ permutation/diagonalization (C/W). Compensation: position-operator systems (quantifier algebraization) vs whole-concept assertion operators (primitive-ization) — two routes. Recycling: γ-representational (FOL = cylindric-algebra equational theory). Pathology: no finite basis for arbitrary recurrence patterns (W/T_f cover special cases only); second-order wall (concept traversal inexpressible — historical patch: type layers).

2.5 Structural interpretation of results

1. **Anchor-previous-station law = single-step moves on the A6 demotion ladder**: blind candidates reach the rung adjacent to the dismantled item (ternary composition / US layer / coordinate-wise operators). Blind sessions perform in-ladder moves (conservative paradigm extension = minimal modification preserving maximal old presuppositions), not D4 framework transfers (paradigm shifts). All three anchor cores require re-filling bridge-class slots — beyond single-step reach.
2. **Error fidelity = cognitive ordering of gear migration**: candidates' "D⇔associativity" overclaim is the pre-1907 E_0-default illusion (undifferentiated "law=structure" intuition); candidates distribute along L4 (feeling the wound), pausing before L5 — reproducing the cognitive order, not a knowledge gap.
3. **H-A2 falsification = slot-nearest-refill principle**: dismantling an operator yields its objectified version (operator-as-object-family), not merely criterion-layer escape — emergence refills the dismantled slot with same-type material. Feedback to theory: the A2/C1 boundary is more permeable than expected (E1-h).
4. **P7 same-lattice = shared 24-slot deep grammar**: the division-of-labor interface cuts exactly between "choosing which slot to operate on" (human problem-sense) and "executing the operation" (machine grammar) — P4: AI never spontaneously chooses law-dismantling (0%); P7: placed in such contexts, AI actions match human history four times over.
5. **B1 = equivalent recurrence is not emergence**: renamed equivalent recurrence fills repair slots with non-triviality zero — true rebuilds must instantiate one of α/β/γ; equivalent recurrence fills no new slot. E1-j elevates this criterion into gate text.
6. **Tri-target same-type emergence**: historical and blind-candidate emergence types match on all three targets (G5 route-switched to the cylindric line but remained α-algebraization) — the strongest blind-condition support for the stage-one law; G5's switch measures the genealogy's width of multi-realization.

§ 3 Experimental Design

Three-zone architecture: Zone A (modern knowledge legal: target selection, freeze-file drafting) → Zone B (T-locked no-foresight: generation/scanning/review) → Zone C (post-round centralized: judgment/scoring/verification), isolated by one-way git-commit gates; Zone-B artifacts never back-edited. **Role separation**: coordinator forbidden to judge or generate; generation, review, judgment in independent sessions; judges blind to review rounds. **Preregistration**: all criteria, weights (repair 40% / PFE-provability 25% / consistency 20% / non-triviality 15%), predictions P1–P8, microhypotheses H-A1/A2 frozen with work-order v1.3.

Target	Facet	Anchors (T_L / T_S)	Span	Candidates
G3′	Law (machine contract)	Hilbert 1899 / Hall 1943	44 yr	4 vendors
G4	Structure (field constraint)	Hausdorff 1914 / Hochster 1969	55 yr	4 vendors
G5	Operator (machine itself)	pre-1924 / Schönfinkel 1924	0 yr	4 vendors

Zone-B pipeline: generation (offline, self-report, descriptive files forbidden) → mechanical era-scan (🟡 marginal / 🔴 definitional-core grading) → PFE four-criterion gate (vocabulary / motivation / mechanism / provability; honest-degradation protection; PASS/FAIL only, no suggestions; grading not elimination). Per-target dual-vendor reviewers; Zone-C: four networked judges per target, four-bin classification (① pre-T_L trivial / ② post-T_S hit 🔥 with ⩾2 rule-level isomorphism points / ③ novel / ④ structural violation).

§ 4 Results

4.1 Gate and scorecard

12 candidates = **5 complete structures + 6 directional drafts + 1 dual-review split** (all drafts retained, G-L5). Pending-item closures: filter 🔴, structural combinator 🔴, W/C notation-layer 🔴. Scorecard: G3′ winner C3-DeepSeek (sole 1.00); G4/G5 three-way ties at 1.00 (ceiling-saturation finding). G1 retrospective calibration (D-18): mechanical order matches expectation → **weights calibrated**. Blind-review discipline evidence: DeepSeek failed its own candidate's PFE-1; Kimi failed its own candidate's PFE-4 — no self-exemption.

4.2 Judgment matrix and the tri-target invariant

Target	①	② 🔥	③	④	Strongest hit
G3′	—	C1/C3/C4 unanimous + C2 split	C2 ×2	—	C3 rule-level (≈ Hall PTR line-by-line); C4 reduction-level (VW 1907 core clause double blind-hit)
G4	1 self-contradictory	C1–C4 majority	C4 ×1	—	C2 (Kolmogorov quotient + blind-proved first-countable $T_2 \Leftrightarrow US$ + cocountable counterexample; three judges independently rank first)
G5	C3 (2/4)	C2/C4 unanimous	C1 ×2	—	C2/C4 (cylindric-algebra parallel: U_i ↔ cylindrification c_i exact)

> **Tri-target invariant**: bin-② majority 10/12 (83%); all three anchor cores zero blind arrivals (S/K 0/16, Moulton 0/16, Hochster 0/16) while all three previous-stations multi-arrived — **reach the station before the anchor, never the anchor**. Dual-primitive flag 0/16: all four G5 candidates flowed into the quantifier-algebraization line, none into the combinator line.

4.3 Hypotheses and control arms

H-A1 (T determines emergence type): **supported** — T1 targets yield object-families, T2 target yields criteria+metalayer, tri-target consistent.
H-A2 (anchor side determines landing): **carrier side fully hit; operator side strict version falsified** — G5 dual landing (operator objectification + metalayer), emergence same-typed as anchor → E1-h refinement.

#	Prediction	Result	Verdict
P1	Criterion discrimination H⩾80% vs R⩽40%	88% vs 10%	**pass**
P2	R-failure concentration trivial+fatal	trivial 13/13 one-sided	**partial**
P3	H band vs R two-tail	held	**pass**
P4	AI signature: named⩾90%, dismantle≈0	100% / 0%	**pass**
P5	Acceptance↔verifier	86% vs 33% (n=31)	**directional**
P6	Six-type directory complete	25/25, zero outside	**pass**
P7	AI actions same-lattice as humans	four replications	**pass**
P8	Degenerate baselines intercepted	B0 pass / B1 partial	**partial**

> **P4 ∧ P7 jointly hold → "humans pose / machines solve" division-of-labor portrait formally unlocked** (D-21 condition met).

4.4 B1 degenerate-baseline live test

Fixture = T_2 renamed "double-neighborhood compatibility condition" reintroduced verbatim (vocabulary airtight). Three blind reviewers: PFE-1/2/3 all PASS, **zero restoration identifications** (ChatGPT alone failed PFE-4 on overclaim); scorecard: non-triviality **0.00** (the only B1-catching dimension), repair-degree 1.00 trap not credited, weighted 0.18 < real-candidate floor 0.25. **Structural finding: the gate's era-faithfulness axis is orthogonal to the cheating axis** — restoring the dismantled condition is precisely the most era-legal content possible; backstop = the 15%-weight non-triviality dimension → E1-j high-priority.

§ 5 Analysis: three headline findings

1. **Conservative-paradigm-extension attractor** (tri-target cross-validated): blind repair intuition starts from operable extensions of the existing paradigm (coordinate operations totalized; convergence behavior stratified; quantifiers algebraized). All three anchor cores require paradigm shifts; none was blindly triggered. Arrival depth tracks task-inherent difficulty, not anchor distance — G5's zero-year span still missed S/K.
2. **Stop-point + error double historical fidelity**: zero non-D model constructions with honest open-problem declaration = the actual 1899 contemporaneous state; candidate errors = pre-1907 folk state / pre-normalization terminology / 1920s hierarchy confusions. Error types are temporally distributed — blind rebuilds "walk the historical road." Cross-reference: the boundaries candidates honestly identified are exactly those history later crossed, by the routes modern review suggested.
3. **Dual-axis decoupling and the division-of-labor portrait**: faithfulness × quality yields three patterns (concentric / decoupled / shifted) — "faithful to the era" and "approaching the historical answer" are separable. P4∧P7: AI never spontaneously dismantles laws (0%) yet matches human historical actions when placed in such contexts (4 replications) — humans pose, machines solve. F7 (honest open-problem declaration) remains an AI-unique behavioral candidate (16/16 candidates co-occur honest-degradation layers; human history shows no publication motivated by open-problem declaration).

§ 6 Honest Disclosure and Limitations (eight items)

1. **Parameter contamination cannot be excluded**: all conclusions phrased as "blind-session rebuilding under explicit knowledge date-locks"; the tri-target invariant (zero anchor-core arrivals) is internal evidence that date-locks suppress voluntary foresight (otherwise anchor cores would be "recalled" like previous stations).

2. **Anchor years are historiographic judgments** (dissent-appendix regime; three fact corrections filed: Moulton qualification, Fano 1892 provenance, Schönfinkel U≠S).
3. **Operationalization flags**: scorecard formula mapping not micro-preregistered (coordinator-drafted, transparently registered); ceiling saturation; G1 calibration evidence tier downgraded; PFE-4 honest-degradation gap; single-letter notation priority; bin-② threshold conceptual-dichotomy (three replications, E1-g).
4. **Sample size**: target-level n=3(+1), candidate-level n=12 — first-round directional records, no extrapolation; no post-hoc subset analysis under anti-mining three-locks.
5. **M8 verifier self-warning**: PFE + judgment sides are themselves filters — non-formalizable targets systematically absent; B1's orthogonal blindness deepens this.
6. **Judge-side boundaries**: structural hit ≠ priority claim; judge-candidate vendor co-sourcing (no self-exemption observed; fully registered); one judge offline-declared (downgraded); one vendor-internal inconsistency (body-text verdict filed).
7. **Process deviations**: G4 dual-review vendor substitution (walled, backfilled, ratified); extra reviewers segregated; B1 over-spec — all two-sided E4 ledger entries (positive side: multi-review discipline's defensive value twice demonstrated).
8. **MR extension status**: Zone-C extension, not preregistered — independently filed, never entering preregistered scoring (dual-axis firewall); from v1.4 slimmed to pure-quality axis (E1-d).

§ 7 Conclusions

1. **Capability**: under date-locks LLM blind rebuilds reliably reach the historical previous-station (83% bin-② hits); anchor cores see zero arrivals — the conservative-paradigm-extension attractor tri-target validated.
2. **Path**: stop points and error morphology both fit historical chronology — structural evidence for blind-rebuild ecological validity.
3. **Division of labor**: humans-pose / machines-solve portrait first-round empirically unlocked (P4∧P7); F7 honest open-problem declaration an AI-unique behavioral candidate.
4. **Methodology**: the three-zone isolation + preregistration + divergence-as-data "controlled historical re-enactment" paradigm is reusable; B1 characterizes the verification system's defense-in-depth; ten E1 entries (a–j) form the v1.4 revision base. The SMPDL structural system simultaneously served as experiment-design language, result-interpretation language, and verification-design basis — one vocabulary through three uses, a first-round demonstration of methodological self-consistency.

Round-Two Supplement • 2026-10-24 (merged in v2.0)

§ R2-1 Design and Execution of Round Two

First full-clause execution of Work Order **v1.4** (E1 ten items + G-L11/12 + P9 iterative-arm preregistration, frozen at commit `150ab55`). Three targets: **G6 commutativity dismantled** (Hamilton 1843 / Frobenius 1877 dual anchors), **G7 phase-volume conservation dismantled** (Poincaré 1890/1899, carrier-side × T3 sole empty grid cell), **G8 unique factorization dismantled** (Kummer 1847 / Dedekind 1871 dual-faced law). Zone-B 12 candidates (Zenodo v3.8 sealed, DOI 10.5281/zenodo.23615249) → full Zone-C chain (dual PFE reviews ×6 + pure-quality-axis MR ×6 + five-band scorecard + dual online judgments ×6 + historiographic comparison + H-A2′ verification + control arms + **P9 three arms**). 22 external sessions; coordinator generated zero content (G-L1).

§ R2-2 Judgment Outcome: First Anchor-Core Hit

TL;DR: G8-C4 ("yin-shu" / factor-bundles) unanimously judged by both adjudicators as a **core hit of the 1871 ideal theory** (seven-row rule-level correspondence table fully isomorphic: set-theoretic definition / operations / inversion / prime=maximal / existence-uniqueness / explicit decompositions (2)=P^2(3)=Q・$\bar{Q}$ / non-principality) — **the first anchor-core arrival in G-series history**. The round-one invariant "arrive at the station before the anchor, never the anchor core" is broken and revised to **"skeleton arrives, foundation stays gapped"**: the lattice canonical-form lemma (sole unproven support) ruled (a)-class — principally repairable within the 1847 toolbox; the ChatGPT judge's incremental finding: the stated condition a | b^2+5 is insufficient (ω-closure forces a | 2b).

§ R2-3 Three Theoretical Updates

1. **Invariant revision**: round-one "station-before-anchor" → round-two "skeleton arrives, foundation stays gapped" (three-layer confluence: scorecard → MR → judgment; anchor-core deliveries co-vary with proof gaps).
2. **Pivot three tiers** (0-year anchor-distance gradient refined): zero-pivot (quaternions = task-prescribed) 4/4 > object revolution (ideals = task-half-hinted) 1/4 > method revolution (measure-theoretic = unhintable) 0/4 direct + 1/4 detour. The type of conceptual revolution the answer requires — not chronological distance — determines blind-arrival rates.
3. **H-A2′ partial support + H-A2″ candidate**: operator-side 2/2 literal fulfillment (dismantled contract rebuilt as object + meta-layer audit); carrier-side first line-switch (G7 criterion-family emergence, object-type 0/4 dual-consensus). Root cause: dismantling a constitutive law yields object-compensators; dismantling a guarantee yields criterion-compensators — **the compensator mirrors the dismantled** (H-A2″, post-hoc construction, triple-locked until round-three preregistration).

§ R2-4 The P9 Contemporary-Review Iterative Arm (three-arm closeout)

Mounted on G8-C4 (judgment-layer sole gap = canonical-form lemma). Ten-step Δ ladder (judgment-round same ruler + γ anchor-post red-line stations) + four counters (retraction / new-construction / **patch** / honest-suspension).

Arm	Input	Δ	Retract/New/Patch/Honest	β2 inversion final state
Main chain (Kimi→DeepSeek→ChatGPT)	Genuine review, 7 items	**1.0**	2/1/4/4→8	Enhanced half-step (universality suspended; theorem claim retracted)
Placebo (Kimi)	Zero mathematical content	0	1/0/0/4→5	Half-step unchanged
Self-assessment (Kimi)	Self-found 5 items	**1.5**	0/0/5/4→4	**Full step (lemma completely proven in-session)**

Three findings: (1) **Cauchy-angle hit** — reviewer DeepSeek self-discovered the b1 lemma gap with zero judgment-layer information (the Liouville 1847-03-01 on-the-spot-challenge role replicated; and incrementally supplied the correct counterexample 2+ω / 2−ω for r1 § 4.2); (2) **patch established as the third modality** — pilot and placebo both retraction-dominant; under genuine review the main chain turns patch-dominant and the self-arm all-patch zero-retraction: the "debridement vs growth" dichotomy is insufficient, H-B stratified by reviser-relation (E1-n); (3) **self 1.5 > main 1.0** (P9-2 control-below-main prediction partially falsified) — reviewer-specificity × reviser-identity two-variable interaction: outsider revisions retract conservatively, family revisions complete directly (same-vendor knowledge continuity; E1-m matrix completion). **Structure-honesty tension**: the same r1 yields "claim landed + lemma complete" (self) vs "claim retracted + lemma suspended" (main) — the deepest is not the most honest (isomorphic to the scorecard robustness-vs-depth decoupling).

§ R2-5 The Dual-Faced Law and Discipline-Scale Emergence (4th data point)

> The hypothesis "discipline-scale emergence ⇔ dual-faced law (T1+T3)" gains its 4th supporting data point: G8 dual-faced → the round's sole anchor-core hit; ideal theory historically grew into the entire discipline of algebraic number theory (known positive); the blind sample's deepest arrival lands exactly on this target. Data points 1–3 (single-faced targets) show no discipline-scale emergence.

§ R2-6 Historiographic Corrections (E1-f annotation protocol)

Yielded by judge online verification: S-1 Kummer 1844 appeared in a Breslau Festschrift (not Crelle 35); S-2 *Les méthodes nouvelles* vol. 1 = 1892 (the Acta-1890 paper anchor unchanged); S-3 quaternion date Oct-16 (all four candidates' uniform Oct-15-night frame = a training-data smoothing trace); S-4 Frobenius Crelle-84 dual 1877/1878 dating; S-5 Kummer-letter dual timelines archived pending publication-grade check.

§ R2-7 Round-Two Conclusions

Round two, executed under the full v1.4 clause set (preregistered scoring formula, ②-threshold bifurcation, restoration-identification, era-scan gate, iterative segregation): the anchor-core zero-arrival invariant is broken and revised to "skeleton arrives, foundation stays gapped"; arrival rates are governed by pivot type rather than chronological distance; the contemporary-review mechanism partially replicates real history (Cauchy-angle hit + isomorphism with the six-week Lamé cycle) and reveals the patch modality and reviser-identity effects. The E1 v1.5 pool (E1-k/l/m/n + S-corrections + H-A2″ triple-locked) registered pending round three. Zenodo v3.9 published; the sequence wall maintained.

附录 · 执行链索引 / Appendix · Execution Chain

Git commits（Zone-C 全链，仓库 SMPDL_GitHub_Pages）：475411b 轮闭合宣告 → 017cef4 PFE 提示词 → a5e20b6 PFE 归档+矩阵 → 2e4d710 MR 提示词 → f48eee7/a662e62/13af8a8 MR 三靶归档+比照 → de59453 分歧存档 → 7763094 评分卡 → 52a3b3a 判定提示词 → a563947/2dbdee1 判定两靶 → 8273355 G3' 判定+第 4 步回验 → 97631fd 对照臂 → 8d57dfd B1 执行卡 → 03cf6a6 B1 终局 → 539b007 E1 登记 → 701ad91 完整试验论文 → 2c43946 结构理论解释篇。
Sealed artifacts: Zenodo v3.6（B 区 12 候选，DOI 10.5281/zenodo.23599041）；v3.7 封装含 Zone-C 全部产物 + 本双语合并论文。
Companion files: G系_完整试验论文 / G系_数学结构理论详细解释 / 各靶判定汇总与历史对照 ×3 / 分歧存档与史学注记 / E1 勘误登记（皆在 reports/g-series/）。

SMPDL G 系 · 双语合并论文 v2.1 · 2026-10-26（术语按 2026-10-26 语言修订注记 M1–M8 对齐：等价复现/相对化/异构替代/未决申报；旧称括注保留） 设计：咖啡 · 协调：史努比（禁判禁生成）· 预注册血统：工单 v1.3（J2） 防挖掘三锁适用 · 体系 F 纯白版式